20 Olly drove 56 km from Liverpool to Manchester.
He then drove 61 km from Manchester to Sheffield.

Olly's average speed from Liverpool to Manchester was 70 km/h.
Olly took 75 minutes to drive from Manchester to Sheffield.

(a) Work out Olly's average speed for his total drive from Liverpool to Sheffield.

First journey: 70 km in 1 hour.

	km	Hours	
×0.8	70	1	×0.8
	56	0.8	

1 mark
Find the first journey time

75 minutes = 75 ÷ 60 = 1.25 hours

Total distance = 56 + 61 = 117 km

Total time = 0.8 + 1.25 = 2.05 hours

1 mark
Add distances or times using consistent units

Average speed = 117 ÷ 2.05 = 57.073... km/h

1 mark
Total distance ÷ total time

1 mark
57 to 57.1 km/h

57.1 km/h

(4)

Janie drove from Barnsley to York.

Janie's average speed from Barnsley to Leeds was 80 km/h.
Her average speed from Leeds to York was 60 km/h.

Janie says that the average speed from Barnsley to York can be found by working out the mean of 80 km/h and 60 km/h.

(b) If Janie is correct, what does this tell you about the two parts of Janie's journey?

The two parts of the journey took equal times.

1 mark
Equal times make the mean of the speeds valid

(1)

(Total for Question 20 is 5 marks)